QML

Quantum Machine Learning

A step-by-step chapter: from gradient descent to a QCNN in CUDA-Q

What you will be able to do after this chapter

- compute losses, gradients, metrics and their statistical uncertainty quickly and reliably;
- work with qubit states, Pauli operators, rotation and controlled gates, tensor products and entanglement;
- encode classical data (basis, angle, amplitude), evaluate quantum kernels and the outputs of small variational circuits;
- derive and apply the parameter-shift rule, and count the cost of exact gradients for shared and controlled parameters;
- read the QCNN of `QML.ipynb` line by line, count its layers, parameters and gates, and interpret its results honestly;
- reason about barren plateaus, noise, shots, simulators, GPUs, classical baselines and the limits of QML;
- solve every type of question in the accompanying 80-question QML assessment.

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Companion material in the same repository: `presentations/QML_Lecture.pdf` (slides), `QML.ipynb` (CUDA-Q notebook) and `assessments/` (question paper and answer book).

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QML — Quantum Machine Learning

How to use this chapter

The chapter is built as a staircase. Sections 1–3 are *classical machine learning and its mathematics*. Sections 4–5 are *quantum computing* (states, measurement, gates, entanglement). Sections 6–10 are *QML itself* (encoding, kernels, variational models, gradients, the QCNN of the notebook). Sections 11–14 are the *advanced* material and the *implementation* (trainability, noise, other algorithms, CUDA-Q, benchmarking). Section 15 collects strategies and a map from every assessment question to the section that teaches it; Section 16 has practice problems with answers.

Statements are marked as *implemented in the repository*, *theory only*, or *illustrative experiment* (run for the lecture with `presentations/qml_source/make_figures.py`). "Previously recorded" results are the outputs stored in `QML.ipynb`; "re-executed" results come from running the same code again (they agree exactly).

Definition: Conventions used everywhere in this chapter

- Qubit 0 is written first: $|q_0 q_1 \ldots\rangle$; two-qubit matrices use the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.
- $R_P(\theta) = e^{-i\theta P/2} = \cos\frac{\theta}{2} I - i \sin\frac{\theta}{2} P$ for $P \in \{X, Y, Z\}$; $R_Y(\theta)\,|0\rangle = \cos\frac{\theta}{2}\,|0\rangle + \sin\frac{\theta}{2}\,|1\rangle$.
- $\langle M\rangle = \langle\psi|M|\psi\rangle$; $\langle Z\rangle = P(0) - P(1)$; Bloch vector $(\langle X\rangle, \langle Y\rangle, \langle Z\rangle)$.
- Labels $y = \pm 1$ with prediction $\operatorname{sign} f$ (or $y \in \{0, 1\}$ with probabilities); MSE $\mathcal{L} = \frac{1}{M}\sum (f_m - y_m)^2$.
- `QML.ipynb`: 2×4 images, 8 qubits, $R_Y(\pi x_i)$ encoding, QCNN with 3 layers $\times$ 4 shared parameters, readout $\langle Z_7\rangle$, COBYLA with 100 loss evaluations, 20 training and 20 test images, seed 42.

1 Learning from data: losses, gradients and neural networks

1.1 The four ingredients

A supervised learning problem has data $(\mathbf{x}_m, y_m)_{m=1}^{M}$, a model $f(\mathbf{x}; \boldsymbol{\theta})$, an optimizer, and a loss

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{M} \sum_{m=1}^{M} \ell(f(\mathbf{x}_m; \boldsymbol{\theta}), y_m).$$

Classification predicts a class, *regression* a number; *unsupervised* learning has no labels (clustering, PCA); *reinforcement* learning learns from rewards.

Definition: common losses

MSE $(f-y)^2$; binary cross-entropy $-[y\ln p+(1-y)\ln(1-p)]$; cross-entropy $-\ln p_{\text{true}}$ with $p_k=e^{z_k}/\sum_j e^{z_j}$; hinge $\max(0,1-yf)$.

1.2 Gradient descent and the chain rule

$\boldsymbol{\theta}\leftarrow\boldsymbol{\theta}-\eta\nabla\mathcal{L}$. For $\mathcal{L}=\frac{1}{2}\boldsymbol{\theta}^{\mathsf{T}}H\boldsymbol{\theta}$ the iteration is $\boldsymbol{\theta}\leftarrow(I-\eta H)\boldsymbol{\theta}$ and converges iff $0<\eta<2/\lambda_{\max}$. Momentum: $v\leftarrow\beta v+g$, $\boldsymbol{\theta}\leftarrow\boldsymbol{\theta}-\eta v$. For linear regression $\nabla\frac{1}{M}\|X\mathbf{w}-\mathbf{y}\|^2=\frac{2}{M}X^{\mathsf{T}}(X\mathbf{w}-\mathbf{y})$.

Worked example 1.1: one step and one neuron

Step 1. $\mathcal{L}=(w_1-3)^2+2(w_2+1)^2$, $\mathbf{w}=(0,0)$, $\eta=0.1$: $\nabla\mathcal{L}=(-6,4)$, new $\mathbf{w}=(0.6,-0.4)$, $\mathcal{L}$: $11\to 6.48$.

Step 2. $f=\sigma(wx+b)$, $L=(f-y)^2$, $x=2$, $w=0.5$, $b=-1$, $y=1$: $z=0$, $f=\frac{1}{2}$, $\partial_w L=2(f-y)\sigma'(z)x=2(-\frac{1}{2})(\frac{1}{4})(2)=-\frac{1}{2}$.

Step 3. Sigmoid + cross-entropy: $\partial L/\partial z=p-y$ (the σ' cancels). A ReLU with negative input passes no gradient.

Key result: activation derivatives

$\sigma'=\sigma(1-\sigma)$, $\tanh'=1-\tanh^2$, $\text{ReLU}'=[z>0]$, $\frac{d}{dz}\ln(1+e^z)=\sigma(z)$. A dense layer $n_{\text{in}}\to n_{\text{out}}$ has $n_{\text{in}}n_{\text{out}}+n_{\text{out}}$ parameters ($8\to 4\to 1$: 41).

Common trap

Moving *along* the gradient; forgetting the factor 2 of a square or the inner derivative x; dividing by $\log_{10}$ instead of ln; summing instead of averaging; using the wrong class's probability in a cross-entropy.

1.3 Non-linearity

XOR is not linearly separable: from $g(0,1),g(1,0)>0$ and $g(0,0),g(1,1)<0$ one gets both $w_1+w_2+2b>0$ and <0. The same argument shows that the clean line images of `QML.ipynb` are not linearly separable: the horizontal pairs in columns 1–2 of both rows light exactly the same four pixels as the vertical lines in columns 1 and 2.

2 Linear algebra and probability for QML

Key result: toolkit

- Eigen-decomposition: PCA directions are eigenvectors of the covariance; explained fraction $\lambda_1/\operatorname{tr}C$.
- Kronecker product $A\otimes B=(a_{ij}B)$; $(A\otimes B)(C\otimes D)=AC\otimes BD$; $\operatorname{tr}(A\otimes B)=\operatorname{tr}A\operatorname{tr}B$; not commutative.
- Kernels: $k(\mathbf{u},\mathbf{v})=e^{-\|\mathbf{u}-\mathbf{v}\|^2/(2\sigma^2)}$; a valid kernel has PSD Gram matrices.
- Bayes: $P(D\mid+)=P(+\mid D)P(D)/P(+)$. Binomial standard error $\sqrt{a(1-a)/M}$.

Worked example 2.2: PCA and Bayes

$C = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$: $\lambda = 3, 1$; first component $(1,1)/\sqrt{2}$ explains 75 %. Recall 0.9, false-positive rate 0.05, prevalence 0.01: $P(D \mid +) = 0.009/0.0585 = 15\,\%$.

3 Evaluating models honestly

Definition: metrics

Accuracy $\frac{TP+TN}{\text{all}}$; precision $\frac{TP}{TP+FP}$; recall $\frac{TP}{TP+FN}$; $F_1 = \frac{2PR}{P+R} = \frac{2TP}{2TP+FP+FN}$.

Method: a trustworthy evaluation

1. split first; fit preprocessing on the training split only; tune on a validation split; touch the test set once;
2. report the test-set size and SE $= \sqrt{a(1-a)/M}$ (20 images, $a = 0.85$: ± 8 points);
3. compare with strong classical baselines on the same data; diagnose gap (overfitting) vs. both-high (underfitting).

Worked example 3.3: the re-executed QCNN on 20 test images

TP 9, FN 1, FP 2, TN 8: accuracy 0.85, precision $9/11 = 0.818$, recall 0.900, $F_1 = 18/21 = 0.857$ (the arithmetic mean 0.859 is *not* F1).

Common trap

Reporting accuracy on imbalanced data; choosing hyper-parameters on the test set; confusing training MSE with accuracy (the recorded run has MSE 0.67 at 90 % accuracy because its outputs are small, $|\langle Z_7 \rangle| \approx 0.2$).

4 Qubits, measurement and the Bloch sphere

$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$, $|\alpha|^2 + |\beta|^2 = 1$. Measurement gives 0 with probability $|\alpha|^2$.

Key result: expectation values of one qubit

$\langle Z \rangle = |\alpha|^2 - |\beta|^2$, $\langle X \rangle = 2\,\mathrm{Re}(\alpha^*\beta)$, $\langle Y \rangle = 2\,\mathrm{Im}(\alpha^*\beta)$. For $\cos\frac{\vartheta}{2}\,|0\rangle + e^{i\varphi}\sin\frac{\vartheta}{2}\,|1\rangle$: $\vec{r} = (\sin\vartheta\cos\varphi, \sin\vartheta\sin\varphi, \cos\vartheta)$.

Worked example 4.4: a phase changes $\langle X \rangle$, not $\langle Z \rangle$

$\frac{1}{2}\,|0\rangle + \frac{\sqrt{3}}{2}e^{i\pi/4}\,|1\rangle$: $\langle Z \rangle = -\frac{1}{2}$, $\langle X \rangle = \frac{\sqrt{3}}{2}\cos\frac{\pi}{4} = \frac{\sqrt{6}}{4}$.

Key result: shots

S shots estimate $\langle Z \rangle$ with standard error $\sqrt{(1-\langle Z \rangle^2)/S}$. 620 zeros in 1000 shots: 0.24 ± 0.031.

Common trap

Forgetting the half angle ($\vartheta = \frac{\pi}{4}$ means amplitudes $\cos\frac{\pi}{8}$, $\sin\frac{\pi}{8}$); treating a global phase as observable; concluding that $R_X(\theta)\,|0\rangle$ and $R_X(-\theta)\,|0\rangle$ are equal because their Z-statistics agree.

5 Gates, tensor products and entanglement

Key result: gate facts

$X^2 = Y^2 = Z^2 = I$; $XY = iZ = -YX$ (cyclic); $(aX+bY+cZ)^2 = (a^2+b^2+c^2)I$. $HZH = X$, $SXS^\dagger = Y$, $R_P(\pi) = -iP$. $R_Y(\theta)^\dagger Z R_Y(\theta) = \cos\theta Z - \sin\theta X$. CNOT: $Z_t \to Z_c Z_t$, $X_t \to X_t$, $X_c \to X_c X_t$. Controlled-$R_Z(\theta) = \mathrm{diag}(1, 1, e^{-i\theta/2}, e^{i\theta/2})$ (control qubit 0).

Definition: product state and entanglement

$\sum c_{ab}\,|ab\rangle$ is a product state iff $c_{00}c_{11} - c_{01}c_{10} = 0$. Purity of the reduced state $\mathrm{tr}\,\rho_1^2 = 1$ (product) down to $\frac{1}{2}$ (maximally entangled).

Worked example 5.5: $R_Y(\theta)$ then CNOT

$\cos\frac{\theta}{2}\,|00\rangle + \sin\frac{\theta}{2}\,|11\rangle$: $\langle Z_1\rangle = \cos\theta$, $\langle Z_0 Z_1\rangle = 1$, $\mathrm{tr}\,\rho_1^2 = \frac{1+\cos^2\theta}{2}$, maximal entanglement at $\theta = \frac{\pi}{2}$. A local $R_Y(\theta)$ on half of $|\Phi^+\rangle$ leaves $\langle Z_1\rangle = 0$ for every θ.

Worked example 5.6: a two-parameter circuit

$R_X(\theta_2)R_Y(\theta_1)\,|0\rangle$: $(0,0,1) \to (\sin\theta_1, 0, \cos\theta_1) \to z = \cos\theta_1\cos\theta_2$. A parameter in a single rotation gate gives $f = a + b\cos\theta + c\sin\theta$ (period 2π).

Key result: memory

n qubits: 2^n amplitudes, $16\cdot 2^n$ bytes (fp64). 30 qubits: 16 GiB; 80 GB GPU: 32 qubits; four of them: 34.

6 Encoding classical data

encoding	state	qubits	gates	remark
basis	$\lvert b_1 \dots b_n\rangle$	1 per bit	$\le n$	binary only
angle	$\bigotimes R_Y(x_i)\,\lvert 0\rangle$	d	d, depth 1	`QML.ipynb`: $R_Y(\pi x_i)$
amplitude	$\sum_j x_j\,\lvert j\rangle\,/\,\lVert\mathbf{x}\rVert$	$\lceil\log_2 d\rceil$	$O(d)$ generic	norm lost
ZZ map	$U_\Phi(\mathbf{x})H^{\otimes n}\cdots$	d	entangling	theory only

Worked example 6.7: the three encodings

Pixel $x = \frac{1}{3}$: $P(1) = \sin^2\frac{\pi}{6} = \frac{1}{4}$, $\langle Z\rangle = \cos\frac{\pi}{3} = \frac{1}{2}$. $\mathbf{x} = (1,2,4,2)$: norm 5, amplitudes $(0.2, 0.4, 0.8, 0.4)$, $P(q_0 = 1) = 0.80$, $P(q_1 = 1) = 0.32$. 256 features of 8 bits: basis 2048, angle 256, amplitude 8 qubits.

Key result: data re-uploading

With L encoding gates $e^{-ixP/2}$, $f(x)$ is a Fourier series with frequencies $0, \dots, L$.

7 Quantum kernels

$k(\mathbf{x}, \mathbf{x}') = |\langle\psi(\mathbf{x})|\psi(\mathbf{x}')\rangle|^2 = \mathrm{tr}[\rho(\mathbf{x})\rho(\mathbf{x}')]$ is symmetric and PSD; estimates from shots may not be. Estimate it by compute–uncompute ($P(0\dots0)$) or by the swap test, $P(0) = \frac{1}{2}(1+k)$.

Key result: the notebook's encoding gives a classical kernel

$k(\mathbf{x}, \mathbf{x}') = \prod_i \cos^2 \frac{\pi(x_i - x_i')}{2}$. Example $\mathbf{x} = (1, 0.5)$, $\mathbf{x}' = (0.5, 0)$: $k = \frac{1}{4}$. For independent uniform features $\mathbb{E}[k] = (\frac{1}{2} + \frac{2}{\pi^2})^n \approx 0.70^n$: kernel values concentrate exponentially.

Worked example 7.8: kernel SVM on the notebook's data (illustrative experiment)

A soft-margin SVM with this kernel, trained on the same 20 images, classifies all 20 test images correctly and 95.2 % of 2000 fresh ones (QCNN: 80.2 %). The kernel was computed from CUDA-Q states and agrees with the formula to 10^{-15} — a classical computation, so this is no quantum advantage. Training needs $\binom{M}{2}$ kernel circuits (190 for $M = 20$).

8 Variational models and hybrid training

$f(\mathbf{x}; \boldsymbol{\theta}) = \langle\psi(\mathbf{x})|U(\boldsymbol{\theta})^\dagger M U(\boldsymbol{\theta})|\psi(\mathbf{x})\rangle$, prediction sign f, loss computed classically, parameters updated by a classical optimizer: *classical input* → *encoding* → *PQC* → *measurement* → *classical loss* → *optimizer* → *update*.

Worked example 8.9: a two-qubit VQC (Heisenberg picture)

$R_Y(x_0) \otimes R_Y(x_1)$, $\mathrm{CNOT}_{0\to1}$, $R_Y(\theta)$ on qubit 1, $f = \langle Z_1\rangle$: $Z_1 \to \cos\theta Z_1 - \sin\theta X_1 \to \cos\theta Z_0 Z_1 - \sin\theta X_1$, so $f = \cos\theta\cos x_0\cos x_1 - \sin\theta\sin x_1$. At $\mathbf{x} = (0, \frac{\pi}{3})$, $\theta = \frac{\pi}{6}$: $f = 0$, $f(\theta \pm \frac{\pi}{2}) = \mp 1$, $f' = -1$, and with $y = 1$: $\partial_\theta (f - y)^2 = 2$.

Key result: cost of training in the notebook

One loss evaluation runs one circuit per training image: $100 \times 20 + 20 = 2020$ circuits, ≈ 1.7 ms each on `qpp-cpu`: ≈ 3.4 s (recorded). MSE 0.505 and accuracy 75 % for outputs $(0.6, -0.2, 0.9, 0.1)$, labels $(+, -, +, -)$.

9 The parameter-shift rule

Key result: two-term rule

For a parameter in exactly one gate $e^{-i\theta P/2}$ ($P^2 = I$): $f = a + b\cos\theta + c\sin\theta$ and $f'(\theta) = \frac{1}{2}[f(\theta + \frac{\pi}{2}) - f(\theta - \frac{\pi}{2})]$ exactly.

Common trap

Shared parameters: $\theta_{1,1}$ of the QCNN appears in 7 gates; shifting it once is wrong — shift every occurrence and add. *Controlled rotations*: generator eigenvalues $0, \pm\frac{1}{2}$; use $f' = d_+[f(\theta+\frac{\pi}{2}) - f(\theta-\frac{\pi}{2})] - d_-[f(\theta+\frac{3\pi}{2}) - f(\theta-\frac{3\pi}{2})]$, $d_\pm = \frac{\sqrt{2}\pm 1}{4\sqrt{2}}$. For $f = \cos\frac{\theta}{2}$ at $\theta = \pi$ the two-term rule gives -0.707, the truth is -0.5.

Key result: cost of one exact QCNN gradient

Per image $4n_{\text{conv}} + 8n_{\text{pool}} = 44 + 56 = 100$ circuits; for 20 images 2000 — the size of the whole COBYLA run.

10 The QCNN of `QML.ipynb`

Definition: layout and blocks

`qcnn_layout(n)`: in each layer, convolution blocks on pairs $(0,1),(2,3),\ldots$ then $(1,2),(3,4),\ldots$ of the active qubits; pooling blocks (source a, sink b) on $(0,1),(2,3),\ldots$; sinks survive; an odd last qubit moves on. Convolution: $\text{CNOT}_{a\to b}(R_Y(\theta_{\ell,1}) \otimes R_Y(\theta_{\ell,2}))$. Pooling: $|1\rangle\langle 1|_a \otimes R_Z(\theta_{\ell,3}) + |0\rangle\langle 0|_a \otimes R_X(\theta_{\ell,4})$ (code: `rz.ctrl; x; rx.ctrl; x`), then a is discarded (deferred measurement).

Key result: counts

Layers $\lceil \log_2 N \rceil$, parameters $4\lceil \log_2 N \rceil$, gates $N + 3n_{\text{conv}} + 4n_{\text{pool}}$. $N = 8$: 11 conv, 7 pool, 69 gates, depth 21, output qubit 7. $N = 16$: 26, 15, 154 gates. $N = 20$: 5 layers, 20 parameters.

Worked example 10.10: the recorded run

CUDA-Q 0.16.0, `qpp-cpu`, no GPU: loss 1.1214 → 0.6704, training accuracy 90 %, test accuracy 85 %, 3.4 s. Re-executed: identical; independent NumPy simulation agrees to 7×10^{-16}.

11 Trainability, shots and noise

Key result: barren plateaus

Deep random circuits: $\text{Var}[\partial_k C] \sim 2^{-n}$, so $\sim 2^n$ shots are needed per gradient component (20 qubits: 10^6). Shallow circuits with local costs: polynomial decay; global costs: exponential. Gradient-free optimizers do not escape. The QCNN is provably free of barren plateaus (Pesah et al. 2021).

Key result: noise

Depolarising noise shrinks $\langle Z \rangle$ roughly by $(1-p)^G$ ($0.99^{69} \approx 0.50$); signs survive longer than margins. Measured (illustrative): the trained QCNN keeps 85 % accuracy up to $p = 0.1$ per gate, but its mean $|\langle Z \rangle|$ falls from 0.171 to 0.003, which would need $\sim 10^5$ shots per image to read.

12 Other QML algorithms (theory only in this repository)

- **QSVM**: classical convex dual with a quantum kernel; $O(M^2)$ kernel circuits; no barren plateaus; concentration risk.
- **QGAN**: generator PQC vs. discriminator; equilibrium value $-\ln 4$.
- **qPCA**: density-matrix exponentiation + phase estimation; needs many copies, low rank, efficient state access; dequantized classically.
- **Clustering**: distances from overlaps, $\|\mathbf{a}-\mathbf{b}\|^2 = 2 - 2\mathbf{a}\cdot\mathbf{b}$; swap test $P(0) = 0.9 \Rightarrow 0.46$.
- **Quantum RL**: PQC policies or Q-functions; Q-learning update $Q \leftarrow Q + \alpha(r + \gamma \max Q' - Q)$ (e.g. $2 \to 3.3$).
- **Quantum data**: the strongest known advantages concern learning from quantum states, not small classical datasets.

13 CUDA-Q, simulation targets and performance

Key result: targets

`qpp-cpu` (recorded run), `nvidia` (one GPU, chosen by `"auto"` if visible), `nvidia+mgpu` (one state vector over several GPUs via MPI), `nvidia+mqpu` (many circuits in parallel — the natural fit for QML), `density-matrix-cpu` (noise, $16 \cdot 4^N$ bytes). `QML.ipynb` uses neither `mgpu` nor `mqpu`; no GPU path was executed.

Worked example 13.11: reading counts

`{00:400, 01:100, 10:300, 11:200}` (qubit 0 first): $\langle Z_0 \rangle = 0$, $\langle Z_1 \rangle = 0.4$, $\langle Z_0 Z_1 \rangle = 0.2$.

Worked example 13.12: time

15 qubits, 0.35 s per circuit (README), 16 images $\times$ 60 evaluations: 960×0.35 s ≈ 5.6 min.

14 Benchmarks and the question of advantage

On the same 20 training images (illustrative experiments): logistic regression on pixels 45 % test (46.5 % on 2000 fresh images — not linearly separable), hand-made rule 100 %, angle-kernel SVM 100 % (95.2 %), QCNN 85 % (80.2 %).

Common trap

Claiming quantum advantage from an accuracy above chance, from beating a weak baseline, or from a kernel that is classically computable. Advantage needs strong baselines, equal budgets, error bars and all costs counted (encoding, shots, optimizer steps).

15 Problem-solving strategies and question map

Method: five habits

1. **Convention first**: which qubit is written first? Which half-angle? Which positive class?
2. **Bloch or Heisenberg**: follow the Bloch vector for one qubit; push observables back through gates for small circuits.
3. **Products factorise**: product states give product kernels and product expectation values.
4. **Count occurrences**, not parameters, for gradients; count blocks from the layout for gates.
5. **Separate** recorded from re-executed from illustrative results; attach an error bar to every accuracy.

Section	Assessment questions
1 Losses, gradients, networks	1, 2, 6, 7, 8, 13, 14, 15, 17, 18, 19, 20
2 Linear algebra, probability	4, 5, 9, 10, 11, 22, 29
3 Evaluation	3, 11, 12, 16, 68, 79, 80
4 Qubits, measurement, Bloch	21, 23, 27, 30, 33, 36, 37
5 Gates, entanglement	24, 25, 26, 28, 31, 32, 34, 35, 38, 39, 40
6 Encoding	41, 42, 43, 54, 55
7 Quantum kernels	9, 44, 45, 46, 58, 70, 74
8 Variational models, hybrid training	34, 52, 53, 56, 59, 60
9 Parameter shift	47, 48, 57, 60, 63, 66
10 QCNN of `QML.ipynb`	48, 49, 50, 51, 53, 66, 68
11 Trainability, shots, noise	26, 27, 61, 62, 64
12 Other algorithms	56, 70, 71, 72, 73, 74
13 CUDA-Q, performance	33, 65, 67, 69, 75, 78
14 Benchmarks, advantage	44, 76, 77, 79, 80

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16 Practice problems

Try each before looking at the answers on the next page. They use the same ideas as the assessment but different numbers.

P1. One gradient step on $\mathcal{L} = 2(w-1)^2$ from $w = 3$ with $\eta = 0.2$: new w and $\mathcal{L}$.

P2. Mean binary cross-entropy for $(p, y) = (0.7, 1)$ and $(0.4, 0)$.

P3. $TP = 30$, $FP = 20$, $FN = 10$: precision, recall, F1.

P4. $\langle Z \rangle$ and $\langle X \rangle$ of $\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$.

P5. Angle-encode the pixel $x = 0.5$ with $R_Y(\pi x)$: $P(1)$ and $\langle Z \rangle$.

P6. Amplitude-encode $(2, 1, 2, 0, 0, 0, 0, 4)$: number of qubits and $P(q_0 = 1)$ (qubit 0 first).

P7. The notebook's kernel for $\mathbf{x} = (0.2, 0.6)$, $\mathbf{x}' = (0.2, 0.1)$.

P8. $f(\theta + \frac{\pi}{2}) = -0.2$, $f(\theta - \frac{\pi}{2}) = 0.6$ for a single $R_Y(\theta)$: $f'(\theta)$.

P9. The notebook's QCNN for $N = 6$: layers, parameters, conv and pool blocks, gates.

P10. Circuits per image for one exact gradient of the $N = 6$ QCNN (two-term per R_Y, four-term per controlled rotation).

P11. Standard error of an accuracy of 0.75 on 40 test images.

P12. Largest fp64 state vector on a 40 GB GPU.

P13. Swap test with $P(0) = 0.6$: the fidelity kernel.

P14. Depolarising factor $(1 - p)^G$ for $p = 0.005$ and the 15-qubit QCNN ($G = 146$).

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Answers to the practice problems

P1. $\mathcal{L}' = 4(w-1) = 8$; $w = 3 - 1.6 = 1.4$; $\mathcal{L} = 2(0.4)^2 = 0.32$.

P2. $\frac{1}{2}(-\ln 0.7 - \ln 0.6) = \frac{1}{2}(0.357 + 0.511) = 0.434$.

P3. $P = 0.6$, $R = 0.75$, $F_1 = 60/90 = 0.667$.

P4. $\langle Z \rangle = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$; $\langle X \rangle = 2 \cdot \frac{\sqrt{3}}{2} \cdot \frac{1}{2} = \frac{\sqrt{3}}{2}$.

P5. $P(1) = \sin^2 \frac{\pi}{4} = \frac{1}{2}$; $\langle Z \rangle = \cos \frac{\pi}{2} = 0$.

P6. Norm 5, 3 qubits; $q_0 = 1$ for indices 4–7: $P = 16/25 = 0.64$.

P7. $\cos^2 0 \cdot \cos^2 \frac{\pi}{4} = \frac{1}{2}$.

P8. $\frac{1}{2}(-0.2 - 0.6) = -0.4$.

P9. $6 \to 3 \to 2 \to 1$: 3 layers, 12 parameters; conv $5 + 2 + 1 = 8$, pool $3 + 1 + 1 = 5$; gates $6 + 24 + 20 = 50$.

P10. $4 \cdot 8 + 8 \cdot 5 = 72$.

P11. $\sqrt{0.75 \cdot 0.25/40} = 0.068$.

P12. $2^{31} \cdot 16 = 34.4$ GB fits, $2^{32} \cdot 16 = 68.7$ GB does not: 31 qubits.

P13. $2(0.6) - 1 = 0.2$.

P14. $0.995^{146} = e^{146 \ln 0.995} \approx 0.48$.

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Summary of the chapter in ten lines

1. Learning = data + model + loss + optimizer; $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla \mathcal{L}$; chain rule for gradients.
2. Evaluate on untouched test data, with error bars ($\sqrt{a(1-a)/M}$) and strong baselines; use P/R/F1 for imbalance.
3. Qubits: $\langle Z \rangle = |\alpha|^2 - |\beta|^2$; half angles; global phases invisible; shots give $\pm 1/\sqrt{S}$.
4. $R_Y^\dagger Z R_Y = \cos\theta Z - \sin\theta X$; CNOT moves $Z_t \to Z_c Z_t$; $\det C = 0 \iff$ product state.
5. Encodings: basis (bits), angle (d qubits, depth 1), amplitude ($\log_2 d$ qubits, deep).
6. Fidelity kernels are PSD; product encodings give classical product kernels that concentrate like 0.70^n.
7. Hybrid loop: encode → PQC → measure → classical loss → optimizer → update.
8. Parameter shift is exact for single Pauli rotations; shared parameters per occurrence; controlled rotations four terms.
9. QCNN: $\lceil \log_2 N \rceil$ layers, 4 shared parameters each; $N = 8$: 12 parameters, 69 gates, 90/85 % recorded.
10. Barren plateaus, noise, shots and data loading limit QML; no quantum advantage is claimed or shown here.